

Bags, Vectors & Transformers

Ethics, Open Models & Distillation · Day 4, Part 2

Denise J. Roth, PhD Candidate

Strategic Communication Group · Wageningen University & Research

The Questions That Matter Most

LLMs are powerful and easy to use. That is exactly why we need to slow down and ask the **harder questions** — especially as researchers, where our choices have consequences for reproducibility, ethics, and trust.

 What are the **ethical and practical risks** of using LLMs in research?

 Why do **open-weight** and **local** models matter so much?

 How can **distillation** give us the best of both worlds?

This is where a thoughtful researcher separates themselves from someone just calling an API.

Risks You Need to Take Seriously

Scientific risks

- **Hallucination:** confident, plausible, wrong
- **Reproducibility:** closed models change or vanish
- **Black box:** you can't inspect how it decided
- **Bias:** absorbed from training data
- **Contamination:** it may have seen your test data

Ethical & practical risks

- **Privacy:** sending sensitive data to a company's servers
- **Consent & GDPR:** is that even allowed?
- **Environmental** cost of large models
- **Dependence** on a commercial provider
- **Cost** that scales with your data

Principles for Responsible Use

Do

- **Validate** against human coding
- **Report** the model, version, prompt, and date
- Check for **bias** in outputs
- Keep a **human in the loop**
- Prefer **transparent, reproducible** setups

Don't

- Send **sensitive / personal data** to a closed API without checking rules
- Treat outputs as **ground truth**
- Hide that you used an LLM
- Assume it will behave the same next month

Transparency is the throughline: another researcher should be able to understand and, ideally, reproduce exactly what you did.

Closed vs. Open-Weight Models

A crucial distinction. **Closed** models (GPT-4, Claude, Gemini) live behind an API. **Open-weight** models (Llama, Mistral, Qwen, Gemma) can be **downloaded and run yourself**.

Closed (API)

- Often the **most capable**
- Easy: just call the API
- But: **data leaves your control**
- Can **change or disappear**
- A **black box** you rent

Open-weight

- **Download** the weights, run locally
- Data **never leaves** your machine
- **Frozen** & reproducible forever
- Free to use, inspect, fine-tune
- Catching up fast on quality

The Case for Open & Local Models

For **research**, open-weight models you run locally are often the *better* choice — not despite being less flashy, but because of what they give you:

Privacy & GDPR

Sensitive data (interviews, patient text) never leaves your control.

🔄 Reproducibility

The exact model is frozen — your study can be rerun in five years.

🔍 Transparency

You can inspect, document, and share exactly what you used.

€ Cost

No per-call fees — run it as much as you like on your data.

🔒 Independence

No lock-in to a company that may change terms or prices.

🔧 Control

Fine-tune it, adapt it, make it *yours*.

*Modern open models run on a single GPU — or even a good laptop — with tools like Ollama.
The barrier is lower than most people think.*

An Honest Caveat

Open & local is not *always* the answer. Being fair about the trade-offs:

Open/local wins when

- Data is **sensitive**
- You need **reproducibility**
- You have **lots** of data to process
- You want full **control**

Closed can win when

- You need the **absolute best** quality
- It is a **one-off**, small task
- Data is **not sensitive**
- You lack local compute

*The point is to **choose deliberately** — not to default to a commercial API just because it's the first thing you saw.*

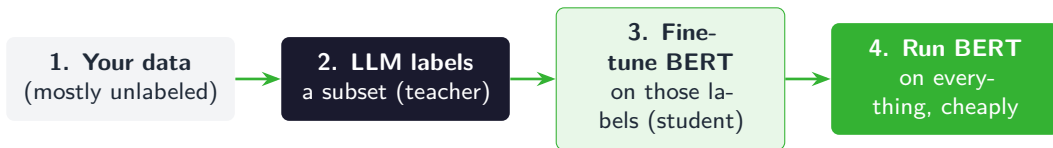
Distillation: The Best of Both Worlds

Here is the elegant idea that ties this whole course together. You do **not** have to choose between a powerful-but-expensive LLM and a small-but-cheap BERT. **Use both.**

The core move: let a big **LLM** label a *subset* of your data, then use those labels to **fine-tune a small BERT-style model** (Day 3). The small model learns to imitate the big one — on your specific task.

This is a form of **knowledge distillation**: transferring what a large “teacher” model knows into a small, efficient “student” model. The student is cheap, fast, local, reproducible — and often nearly as good *on your task*.

The Distillation Workflow



You pay the LLM cost **once**, for a subset. The resulting BERT model then labels your **entire** corpus — and every future corpus — for almost nothing, on your own machine.

Why Distillation Is So Attractive for Research

You get. . .

- LLM-level **quality** on your task
- BERT-level **speed & cost**
- A **local, reproducible** final model
- No sensitive data sent to APIs at scale
- A model you **own** and can share

Keep in mind

- The student inherits the teacher's **biases & errors**
- Still **validate** against human labels
- Needs enough labeled examples to learn from
- Works best for **well-defined** tasks

*This is arguably the **sweet spot** for computational social science right now: LLM power, BERT practicality, full transparency.*

The Whole Journey

Four days, one arc — and it all comes together in that last idea:



The newest tools do not make the older ones obsolete. A dictionary, a TF-IDF classifier, a fine-tuned BERT, an LLM annotator — each has its place. **The skill is choosing well.**

Start simple, add complexity only when it earns its place, validate everything, and stay transparent. That is computational text analysis done well.

Thank you!

From bags to vectors to transformers to LLMs — you now have the full modern toolkit for computational text analysis.